

CHRISTOPHER DAVENPORT

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SELECTED ANALYTICS PROJECTS

SQL Analytics on a Relational Retail Database | MySQL, Sakila - [Link](#)

- Answered multi-part business questions across a 16-table normalized schema, including top-grossing film category per country and highest-spending customer per store, using chained CTEs, window functions (ROW_NUMBER, RANK, LAG, running totals), and nine-table join chains.
- Diagnosed a join-path error that would have misattributed revenue, tracing transactions through the inventory table (the physical copy tied to a store) rather than the customer's registered store, after verifying that customers transact across both locations.
- Applied NULL-safe logic throughout, using NOT EXISTS over NOT IN to avoid three-valued-logic failures and CASE handling to prevent NULL propagation through percentage calculations.
- Validated all table row counts against canonical values at load, documenting a discrepancy as a source-file issue rather than a load error before any analysis was run.

Datacenter REIT Market Stability Analysis | Python, MySQL, yfinance, Excel, Power BI - [Link](#)

- Designed a normalized MySQL star schema with four relational tables tracking 12 months of daily price data across three datacenter REITs (Equinix, Digital Realty, Iron Mountain), totaling 753 records, and transformed raw daily prices into monthly volatility and growth metrics.
- Built an end-to-end ETL pipeline in Python with yfinance and SQLAlchemy, and ran per-company linear regression in statsmodels, identifying autocorrelation as an assumption violation and reporting results descriptively rather than overstating statistical significance.
- Developed interactive Power BI dashboards using DAX measures, DirectQuery, and dynamic period-comparison logic, plus an Excel pivot dashboard, to serve both technical and non-technical audiences.
- Engineered a real-time data pipeline using yfinance's asynchronous WebSocket API with concurrent scheduled polling, writing live prices to MySQL on independent refresh intervals.

Teen Social Media & Physical Activity Analysis | Python, pandas, SQL, SQLite, statsmodels, Power BI - [Link](#)

- Built an end-to-end analytics workflow using Python, pandas, SQLite and Power BI to examine relationships among teen sleep, bedtime screen time and physical activity.
- Imported a structured Kaggle dataset into SQLite, queried analysis variables and recoded categorical data for regression modeling.
- Conducted OLS regression and moderation analysis in Python using statsmodels, evaluating assumptions with descriptive statistics, visual diagnostics, correlation matrices, VIF and tolerance.
- Exported regression outputs to SQLite and wrote SQL views to format model coefficients, descriptive statistics and grouped metrics for dashboard reporting.
- Designed a two-page Power BI report with separate technical and non-technical dashboards to communicate model results, descriptive trends and key behavioral metrics.

CORE SKILLS

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|---|--|
| • Data Cleaning and Data Wrangling | • Statistical Analysis and Regression Modeling |
| • Advanced SQL and Relational Data Modeling | • Data Visualization and Dashboard Reporting |
| • Data Acquisition and Exploratory Analysis | • Cross-Disciplinary Analytic Problem-Solving |

CERTIFICATIONS

Databases and SQL for Data Science with Python, IBM/Coursera	July 2026
Data Analysis with Python, IBM/Coursera	August 2026
Python for Data Science, AI & Development, IBM/Coursera	July 2026
Data Analyst Professional Certificate, IBM/Coursera	Expected August 2026

EXPERIENCE

Mercor Intelligence

June 2026 – August 2026

Independent Contractor

- Providing expertise in quality assurance for a top AI lab.

Close Relationships and Health Lab, The University of Texas at Dallas

August 2024 – July 2026

Principal Investigator and Lead Data Analyst

- Built an end-to-end data processing pipeline in R for 3+ active datasets, performing data cleaning, variable structuring and quality checks using distributions and descriptive statistics.
- Designed and executed an independent behavioral research study on interpersonal conflict, using gaps in prior literature to develop research questions, hypotheses and study procedures.
- Developed and executed a mediation analysis plan using structural equation modeling to test hypothesized relationships among study variables.
- Synthesized and presented research findings to both academic and general audiences, translating complex statistical results into clear and accessible narratives.

Close Relationships and Health Lab, The University of Texas at Dallas

March 2024 – June 2026

Research Assistant, Data Manager

- Retrieved, processed, and validated data from multiple sources across audio and GPS collection pipelines, ensuring proper documentation and procedural compliance before analysis.
- Reconstructed legacy datasets with incomplete documentation by reverse-engineering relationships between raw and derived files to restore a functional analysis pipeline.
- Coordinated with stakeholders to identify and resolve missing or misallocated data, improving dataset completeness and reliability.

Couples' Daily Lives Lab, The University of Texas at Dallas

January 2024 - Present

Research Assistant, Behavioral Coder and Quantitative Analyst

- Co-developed and maintained the lab's data dictionary, standardizing variable definitions across 3+ concurrent research projects and reducing interpretive ambiguity.
- Translated observed behaviors into structured quantitative variables through systematic coding procedures, producing analysis-ready datasets from unstructured source material.
- Led calibration sessions for a team of 4+ coders, resolving discrepancies and maintaining measurement consistency across 2+ years of data collection.

EDUCATION

The University of Texas at Dallas

May 2026

M.S., Psychological Sciences

GPA: 4.00

Phi Kappa Phi Honor Society

The University of Texas at Dallas

May 2023

B.S., Psychology, Magna Cum Laude

GPA: 3.89

PUBLICATIONS AND PRESENTATIONS

Davenport, C., et al. (February 2025). Simmer down: Emotional volatility, interpersonal mindfulness, and couples' post-conflict recoveries. Society for Personality and Social Psychology Annual Convention, Denver, CO. [Lead Author and Presenter]

ADDITIONAL INFORMATION

Statistical Methods: ANOVA, Multiple Regression, Moderation Analysis, Mediation Analysis, Multilevel Modeling, Structural Equation Modeling (SEM), Descriptive Statistics, Hypothesis Testing

Tools: SQL, MySQL, SQLite, Python, R, Excel, Power BI, Jupyter Notebook, Visual Studio Code, SPSS, IBM Cognos Analytics